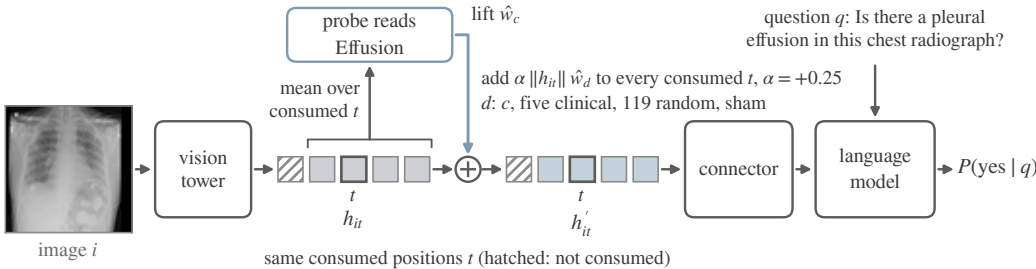


(a) read and write at the final visual block the language model consumes



(b) the grade: one write, three comparisons

$W_{q,d}$: mean change in $P(\text{yes} | q)$ under the write of d

- own effect
 $W_{q,q} > 0$ ✓
- random and sham
 $W_{q,q} > \text{random p95 and } > \text{sham}$ ✓
- five clinical directions
 $W_{q,q} - W_{q,d} > 0$ for every $d \neq q$ ✗

owned: all three hold under simultaneous max- T bounds
seed cell: stronger competitor

(c) seed cell: Effusion, Qwen2.5-VL-7B, NIH

